

VIRTUALIZATION USING PYTHON

INTERNSHIP REPORT

Submitted by

BATCH-7

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In partial fulfillment of the requirement for the award of

**DIPLOMA IN INFORMATION
TECHNOLOGY**

STATE BOARD OF TECHNICAL EDUCATION

GOVERNMENT OF TAMILNADU



JUNE 2023

DEPARTMENT OF INFORMATION TECHNOLOGY

PSG POLYTECHNIC COLLEGE

(Autonomous and an ISO 9001 certified Institution)

COIMBATORE – 641 004

PSG POLYTECHNIC COLLEGE

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**DEPARTMENT OF INFORMATION
TECHNOLOGY**

COIMBATORE – 641 004

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during the academic year

2023

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Faculty Guide

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PSG POLYTECHNIC COLLEGE: COIMBATORE-04
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Self-assessment of Internship by the student

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3. Guide from Industry : Mr. Praveen
4. Date of commencement : 10.05.23-03.06.23
5. Number of day's present : 21 days out of 21 days.

I hereby declare that; I have learned the following skills during my Internship:

S. No	Description	Weight age (%)
1	PYTHON	60%
2	HTML	30%
3	CSS	10%

Date:(/06/2023)

Signature of Student

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5. Number of day's present : 21 days out of 21 days.

I hereby declare that; I have learned the following skills during my Internship:

S. No	Description	Weight age (%)
1	PYTHON	70%
2	HTML	15%
3	CSS	15%

Date:(/06/2023)

Signature of Student

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5. Number of day's present : 21 days out of 21 days.

I hereby declare that; I have learnt the following skills during my Internship:

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3	CSS	05%

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Signature of Student

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5. Number of day's present : 21 days out of 21 days.

I hereby declare that; I have learnt the following skills during my Internship:

S. No	Description	Weight age (%)
1	PYTHON	55%
2	HTML	35%
3	CSS	10%

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Date: 03/06/2023

TO WHOMSOEVER IT MAY CONCERN

This is to certify that Ms. ABHIJITHA K, (Reg. No.21DI01) a student of Second year Diploma in INFORMATION TECHNOLOGY from PSG Polytechnic College, Coimbatore has been successfully completed her Internship Training from 10-May 2023 to 3-June 2023 under a domain of "VIRTUALIZATION". This Training was coordinated and supervised by Blessington Praveen software developer, crapersoft.

We wish Ms. ABHIJITHA K all the best in her future endeavors.

Thanking You

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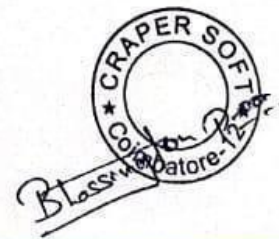
TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Mr. KIRAN T S, (Reg. No. 21DI14)** a student of Second year **Diploma in INFORMATION TECHNOLOGY** from **PSG Polytechnic College, Coimbatore** has been successfully completed his Internship Training from **10-May 2023 to 3-June 2023** under a domain of **"VIRTUALIZATION"**. This Training was coordinated and supervised by **Blessington Praveen** software developer, crapersoft.

We wish **Mr. KIRAN T S** all the best in his future endeavors.

Thanking You

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This is to certify that Mr. LAKSHMAN DEV V K, (Reg. No. 21DI15) a student of Second year Diploma in INFORMATION TECHNOLOGY from PSG Polytechnic College, Coimbatore has been successfully completed his Internship Training from 10-May 2023 to 3-June 2023 under a domain of "VIRTUALIZATION". This Training was coordinated and supervised by Blessington Praveen software developer, crapersoft.

We wish Mr. LAKSHMAN DEV V K all the best in his future endeavors.

Thanking You

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TO WHOMSOEVER IT MAY CONCERN

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We wish **Mr. RAJARAJAN C** all the best in his future endeavors.

Thanking You

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PSG POLYTECHNIC COLLEGE

DEPARTMENT OF INFORMATION

TECHNOLOGY

CANDIDATE'S DECLARATION

I “***ABHIJITHA.K***” hereby declare that I have undertaken three weeks Internship *at* “***CRAPER SOFT***” for a period of 21 days in partial fulfillment of requirements for the award of a Diploma in Information Technology at PSG Polytechnic College. The work which is being presented in the internship report submitted to the Department of Information Technology at PSG Polytechnic College is an authentic record of internship work.

Signature of the Student

(Name: ABHIJITHA.K)

(Reg.no: 21DI01)

PSG POLYTECHNIC COLLEGE

DEPARTMENT OF INFORMATION

TECHNOLOGY

CANDIDATE'S DECLARATION

I ***“KIRAN.TS”*** hereby declare that I have undertaken three weeks Internship *at “CRAPER SOFT”* for a period of 21 days in partial fulfillment of requirements for the award of a Diploma in Information Technology at PSG Polytechnic College. The work which is being presented in the internship report submitted to the Department of Information Technology at PSG Polytechnic College is an authentic record of internship work.

Signature of the Student

(Name: KIRAN.TS)

(Reg.no: 21DI14)

PSG POLYTECHNIC COLLEGE

DEPARTMENT OF INFORMATION

TECHNOLOGY

CANDIDATE'S DECLARATION

I ***“LAKSHMANDEV.VK”*** hereby declare that I have undertaken three weeks Internship *at “CRAPER SOFT”* for a period of 21 days in partial fulfillment of requirements for the award of a Diploma in Information Technology at PSG Polytechnic College. The work which is being presented in the internship report submitted to the Department of Information Technology at PSG Polytechnic College is an authentic record of internship work.

Signature of the Student

(Name: LAKSHMANDEV.VK)

(Reg.no: 21DI15)

PSG POLYTECHNIC COLLEGE
DEPARTMENT OF INFORMATION
TECHNOLOGY

CANDIDATE'S DECLARATION

I ***“RAJARAJAN.C”*** hereby declare that I have undertaken three weeks Internship ***at “CRAPER SOFT”*** for a period of 21 days in partial fulfillment of requirements for the award of a Diploma in Information Technology at PSG Polytechnic College. The work which is being presented in the internship report submitted to the Department of Information Technology at PSG Polytechnic College is an authentic record of internship work.

Signature of the Student

(Name: RAJARAJAN.C)

(Reg.no: 21IH12)

ABSTRACT

ON

VIRTUALIZATION

Virtualization is defined as a framework or methodology of dividing computer hardware resources into multiple execution environments, by applying one or more concepts or technologies such as hardware and software partitioning, time-sharing, partial or complete machine simulation, emulation, quality of service, and many others. The objectives of any virtualization technology include adding a layer of abstraction between the applications and the hardware, reducing costs and complexity, isolating computer resources for improved reliability and security, improving service levels and the quality of service, etc. Software, referred to as a virtual machine monitor (VMM), controls the use and access to the CPU, memory, storage, and network resources underneath. Server virtualization is the more dominant form of virtualization in use today. Implementations of server virtualization exist on, and for all, CPU platforms and architectures, the most popular being the IA-32 or x86. In its most basic form, storage virtualization exists in assembling multiple physical disk drives, or spindles, into a single entity presented to the host server and operating system, such as with RAID implementations. The true storage tier and its components were further masked by introducing and adopting storage area network (SAN) technologies. The most popular forms of network virtualization are virtual LAN (VLAN), virtual IP (VIP), and virtual private network (VPN). Application virtualization, or software virtualization, is the newest member of the ever-growing virtualization family. It is a new approach to software management.



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We have taken efforts in this project. However, it would not have been possible without the kind support and help of many individuals. We would like to extend our sincere thanks to all of them.

Any omission in this brief acknowledgment does not mean a lack of gratitude.

About the Company



CRAPER SOFT is a newly developed Global Technology Development Corporation, building new technical skilled manpower for global industry requirements. The company was set up in 2015, to help nascent Engineering Pupils and industry clients overcome their Technological challenges. It has today grown up to be amongst the country's leading talent development companies offering optimal solutions to Individuals, Enterprises, and Institutions across the country.

CRAPER SOFT provides real-time product solution in Engineering and Technology; also train professional life skill and touches countless learners every year.

CRAPER SOFT'S expertise in learning content-based development and training delivery; Implementing Advanced Technology and educating process management make it the most preferred training partner across the country

Research-based Innovation, a key driver at CRAPER SOFT, has enabled the organization to develop programs and curricula that use cutting-edge industrial design methodologies and training delivery. CRAPER SOFT Individual Learning Business offerings include industry-endorsed programs for students seeking careers in various MNCs and also solve the problem in time in an efficient manner

Research of the company:

Innovation has been and will always be the foundation of CRAPER SOFT. The objective is to: To create ideas for changing times. Ideas that fully realize the potential of technology. Ideas that understand the different needs of people from different parts of the world.

The brand has fresh thinking with a modern outlook and is ever-adapting to customers' needs to enrich across the globe with its cutting-edge technology and innovation-driven approach. CRAPER SOFT provides a wide range of Tech products such as Microcontroller Development Boards, DSP Boards, CPLD /FPGA Boards, Communication kits (ZIGBEE, GSM, GPS, and RF Modules), Power Electronics kit, Basic Electronics Items, Sensors, and much more.

We ensure the performance of the goods will be high quality and the price is affordable

CHAPTER 1

INTRODUCTION

Virtualization is the technology that allows the creation of virtual environments or machines within a physical infrastructure, providing benefits such as resource optimization, flexibility, security, and efficient management of computing resources. It enables the abstraction and isolation of software from the underlying hardware, revolutionizing the way IT infrastructure is utilized and managed



Figure 1.1 Introduction

1.1 Definition

Virtualization is the creation of virtual instances that mimic physical computing resources, enabling efficient utilization, flexibility, and simplified management of IT infrastructure.

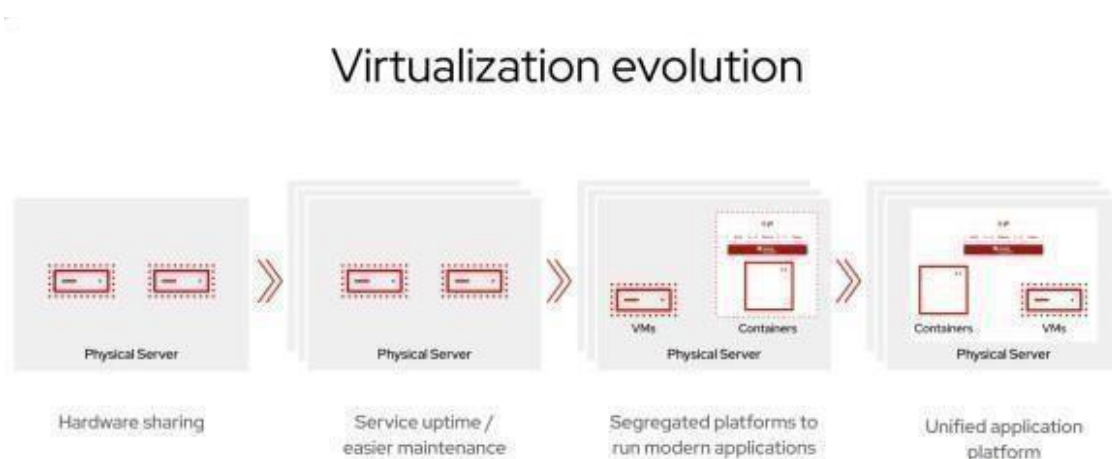


Figure 1.2 Evolution

1.2 Brief History and Evolution of Virtualization

1. Hypervisor-Based Virtualization (1980s-1990s):

The development of hypervisors, or virtual machine monitors (VMMs), introduced a new level of virtualization. Hypervisors allow multiple operating systems to run concurrently on a single physical machine. IBM's VM/370 and later VMware's VMware Workstation were notable hypervisor-based virtualization solutions.

2. Operating System-Level Virtualization (late 1990s-2000s):

Operating system-level virtualization, also known as containerization or OS virtualization, emerged. It allows multiple isolated user-space instances, called containers, to run on a shared operating system kernel. Technologies like FreeBSD Jails, Solaris Zones, and later Docker popularized this lightweight form of virtualization.

3. Hardware-Level Virtualization (2000s-present):

Hardware-level virtualization, powered by Intel VT-x and AMD-V hardware extensions, became widespread. It enables efficient and secure virtualization of CPU, memory, storage, and networking resources. Products like VMware ESXi, Microsoft Hyper-V, and open-source solutions like KVM and Xen utilize hardware-level virtualization.

4. Cloud Virtualization (2000s-present):

The advent of cloud computing brought virtualization to a broader audience. Cloud service providers leverage virtualization technologies to offer users scalable, on-demand virtual resources. Platforms like Amazon EC2, Microsoft Azure, and Google Cloud heavily rely on virtualization for their infrastructure.

5. Software-Defined Infrastructure (SDI):

Virtualization forms the foundation of the software-defined infrastructure concept. SDI enables the abstraction and virtualization of all data center resources, including computing, storage, and networking. It allows for dynamic, automated management of resources through software control and orchestration.

1.3 Importance and Relevance of Virtualization in the modern computing landscape

Virtualization plays a crucial role in the modern computing landscape and has become a fundamental technology for businesses and organizations. Here are some key reasons why virtualization is important and relevant today:

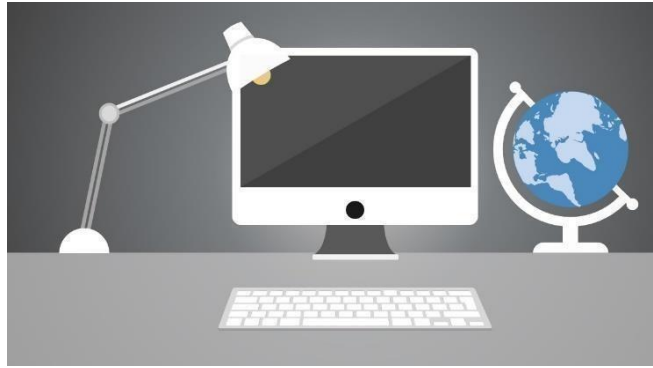


Figure 1.3 Modern PC

1. **Resource Optimization:**

Virtualization allows for efficient utilization of hardware resources by running multiple virtual machines (VMs) on a single physical server. This leads to cost savings, reduced energy consumption, and a smaller physical footprint in data centers.
2. **Scalability and Flexibility:**

Virtualization enables easy scalability of computing resources. Virtual machines can be quickly provisioned, cloned, or scaled up/down to meet changing demands. It provides the flexibility to allocate resources based on workload requirements and ensures efficient resource utilization.
3. **Cost Savings:**

By consolidating multiple servers onto a single physical machine through virtualization, organizations can reduce hardware costs, maintenance expenses, and power consumption. Virtualization also enables better server management and simplifies software licensing.
4. **Simplified Management:**

Virtualization simplifies the management of IT infrastructure. It provides centralized control and administration of virtual machines, making tasks such as provisioning, monitoring, and backup/recovery more streamlined and efficient.

5. Improved Disaster Recovery:

Virtualization facilitates effective disaster recovery strategies. VM snapshots, replication, and migration capabilities enable faster backup and recovery processes. Virtual machines can be easily moved between physical servers, ensuring high availability, and reducing downtime in case of hardware failures or system maintenance.

6. Enhanced Security and Isolation:

Virtual machines provide isolation between applications and operating systems running on the same physical server. Security vulnerabilities within one virtual machine are contained and do not impact others, making virtualization a valuable tool for enhancing system security.

7. Legacy System Support:

Virtualization allows organizations to run legacy applications and operating systems on modern hardware. It provides a bridge between older software and new infrastructure, extending the lifespan of legacy systems and ensuring compatibility.

8. DevOps and Testing:

Virtualization supports agile software development and testing processes. Virtual machines can be used to create isolated testing environments, enabling developers to test applications across different operating systems and configurations without impacting production systems.

9. Cloud Computing:

Virtualization forms the foundation of cloud computing. Cloud service providers leverage virtualization technologies to deliver on-demand resources, elasticity, and scalability to users. Virtualization enables the efficient and secure sharing of physical infrastructure among multiple users in a cloud environment

CHAPTER – 2

Types Of Virtualizations

There are several types of virtualizations, each serving different purposes and utilizing different levels of abstraction. The main types of virtualizations include:



Figure 2.1 Types

2.1 Server Virtualization:

Server virtualization involves dividing a physical server into multiple virtual machines (VMs), each running its own operating system and applications. It enables efficient utilization of server resources, consolidation of multiple servers onto a single physical host, and improved flexibility in resource allocation.

2.2 Desktop Virtualization:

Desktop virtualization allows multiple virtual desktop instances to run on a single physical machine, either locally or remotely accessed by users. It provides centralized management, security, and flexibility in delivering desktop environments to users, enabling easy provisioning, maintenance, and updates.

2.3 Application Virtualization:

Application virtualization separates applications from the underlying operating system, allowing them to run in an isolated virtual environment. It eliminates application conflicts, simplifies deployment, and enables applications to run on different operating systems without modification.

2.4 Network Virtualization:

Network virtualization abstracts the physical network infrastructure, allowing the creation of virtual networks on top of it. It enables logical segmentation, isolation, and customization of network resources, improving flexibility, scalability, and management of network infrastructure

2.5 Storage Virtualization:

Storage virtualization abstracts physical storage resources and presents them as logical volumes or virtual disks. It allows for centralized storage management, simplifies provisioning, migration, and backup processes, and provides features such as data deduplication and thin provisioning.

2.6 Operating System Virtualization/Containerization:

Operating system virtualization, also known as containerization or OS-level virtualization, allows multiple isolated user-space instances, called containers, to run on a shared operating system kernel. It provides lightweight and efficient virtualization, with minimal performance overhead, ideal for deploying and managing applications in a fast and scalable manner.

2.7 Hardware Virtualization:

Hardware virtualization utilizes hardware-assisted virtualization technologies, such as Intel VT-x or AMD-V, to enable efficient and secure virtualization of CPU, memory, storage, and networking resources. It forms the basis for other virtualization types and allows for the creation of virtual machines that closely resemble physical machines

CHAPTER – 3

Challenges and Consideration

While virtualization offers numerous benefits, there are also several challenges and considerations that organizations should be aware of. These include:



Figure 2.1 Challenges

3.1 Performance Overhead:

Virtualization introduces an additional layer of software (hypervisor) between the hardware and the virtual machines, which can result in performance overhead. While the performance impact has significantly With a deep reverence, we acknowledge loads may still experience a slight decrease in performance compared to running on bare-metal hardware.

3.2 Resource Allocation and Management:

Efficient resource allocation and management are crucial in virtualized environments. Organizations need to carefully plan and allocate CPU, memory, storage, and network resources to ensure optimal performance and avoid resource contention. Effective monitoring, capacity planning, and resource management tools are essential to maintain performance and prevent resource bottlenecks.

3.3 Security and Isolation:

Ensuring security and isolation between virtual machines is critical to protect sensitive data and prevent unauthorized access. While virtualization provides isolation at the software level, vulnerabilities in the hypervisor or misconfigurations can potentially compromise the security of all virtual machines running on the same host. Strong security measures, regular patching, and strict access controls are necessary to mitigate these risks.

3.4 Virtual Machine Sprawl:

Virtualization makes creating and deploying virtual machines easier, which can lead to virtual machine sprawl. Without proper management and monitoring, the proliferation of virtual machines can result in increased management complexity, resource wastage, and higher operational costs. Implementing policies and practices for VM lifecycle management, including provisioning, decommissioning, and monitoring, is essential to avoid sprawl.

3.5 Backup and Disaster Recovery:

Virtualized environments require robust backup and disaster recovery strategies. Traditional backup approaches may not be suitable for virtual machines, as they may not capture the entire system state. Specialized backup solutions that integrate with the virtualization platform, such as image-based backups and snapshots, are often necessary. Additionally, organizations should have well-defined disaster recovery plans to ensure business continuity in case of virtual infrastructure failures.

3.6 Licensing and Compliance:

Virtualization can introduce licensing and compliance challenges, especially when running software that is licensed based on physical hardware or core count. Virtual machines can dynamically migrate between hosts, making it difficult to track and manage license usage. Organizations need to carefully review software licensing agreements and ensure compliance with licensing terms in virtualized environments.

3.7 Training and Skillset:

Virtualization technologies often require specialized skills and knowledge to implement. Many individuals' kind support and help IT staff need to be trained and up-to-date with virtualization technologies, including hypervisors, management tools, and best practices. Investing in training and professional certifications can help organizations overcome skillset challenges and maximize the benefits of virtualization.

CHAPTER-4

Benefits and Advantages of Virtualization

Virtualization offers numerous benefits and advantages in the realm of computing and IT infrastructure. Here are some key advantages of virtualization:



Figure 4.1 Benefits

4.1 Resource Optimization:

Virtualization allows for better utilization of computing resources. By running multiple virtual machines (VMs) on a single physical server, organizations can maximize the use of CPU, memory, storage, and network resources. This leads to cost savings and increased efficiency.

4.2 Cost Reduction:

Virtualization helps reduce hardware and operational costs. By consolidating multiple servers onto a single physical machine, organizations can minimize hardware purchases, power consumption, cooling costs, and physical space requirements in data centers. Additionally, virtualization streamlines management tasks, reducing administrative overhead and labor costs.

4.3 Improved Flexibility and Scalability:

Virtualization provides resource allocation and scaling flexibility. Virtual machines can be easily provisioned, cloned, or migrated, allowing for dynamic adjustment of computing resources based on demand. This scalability ensures that resources can be efficiently allocated to meet changing workload requirements.

4.4 Simplified Management:

Virtualization simplifies the management of IT infrastructure. Administrators can centrally manage and control multiple virtual machines from a single management interface, reducing complexity and improving operational efficiency. Tasks such as provisioning, backup, recovery, and software updates can be automated and streamlined.

4.5 Enhanced Disaster Recovery and Business Continuity:

Virtualization offers robust disaster recovery capabilities. Virtual machines can be easily backed up, replicated, and migrated, making it simpler to recover from hardware failures or system crashes. Virtualization also enables the creation of redundant and highly available configurations, ensuring business continuity and minimizing downtime.

4.6 Increased Security and Isolation:

Virtualization provides enhanced security and isolation between different virtual machines. Each virtual machine operates in its isolated environment, isolating applications, and operating systems from each other. This isolation helps prevent security breaches and contains the impact of any vulnerabilities or attacks.

4.7 Testing and Development Efficiency:

Virtualization is valuable for testing and development environments. Virtual machines can be quickly created, cloned, or reverted to a previous state, making it easy to set up isolated test environments and accelerate software development cycles. Virtualization enables developers to test applications on different operating systems and configurations without the need for separate physical hardware.

4.8 Legacy System Support:

Virtualization allows organizations to run legacy applications or operating systems on modern hardware. By encapsulating legacy systems within virtual machines, businesses can extend the lifespan of older software and leverage newer infrastructure without compatibility issues.

4.9 Cloud Computing Enablement:

Virtualization is a foundational technology for cloud computing. Cloud service providers utilize virtualization to offer scalable and on-demand resources to users, enabling the efficient sharing of physical infrastructure among multiple tenants. Virtualization enables the flexibility and elasticity required for cloud deployment.

CHAPTER-5

Virtualization in cloud computing

Virtualization is a foundational technology in cloud computing and plays a vital role in enabling cloud environments' key characteristics and benefits. Here's how virtualization is closely tied to cloud computing:



Figure 5.1 CC

5.1 Multi-Tenancy and Resource Sharing:

Virtualization allows for the efficient sharing of physical computing resources among multiple tenants in a cloud environment. Virtual machines (VMs) can be created on a shared physical infrastructure, providing isolated and dedicated computing environments for each tenant while maximizing resource utilization.

5.2 Scalability and Elasticity:

Virtualization enables the scalability and elasticity of cloud computing. With virtualization, organizations can quickly provision and deploy new VMs, clone existing ones, or scale up/down resources as needed to meet fluctuating workload demands. This dynamic resource allocation allows for rapid scalability and ensures optimal resource utilization.

5.3 Resource Pooling and Management:

Virtualization enables the pooling and management of computing resources in a cloud environment. Physical servers can be abstracted into virtualization hosts, creating a resource pool that can be efficiently managed and allocated to VMs based on demand. Virtualization management tools provide centralized control and monitoring capabilities for resource allocation and utilization.

5.4 Agility and Flexibility:

Virtualization enhances the agility and flexibility of cloud computing platforms. Virtual machines can be easily migrated between physical hosts without disrupting services, enabling workload balancing, maintenance, and improved availability. This flexibility allows for the efficient allocation of

resources and better responsiveness to changing business needs.

5.5 Rapid Deployment and Provisioning:

Virtualization simplifies the deployment and provisioning of new computing instances in the cloud. VMs can be rapidly created from pre-configured templates, enabling faster application deployment and reducing the time required for setting up and configuring hardware resources.

5.6 Isolation and Security:

Virtualization provides strong isolation between VMs in a cloud environment, enhancing security and ensuring that workloads are isolated from each other. Each VM operates in its own virtualized environment, isolating the operating system, applications, and data from other VMs on the same physical infrastructure. This isolation helps prevent the spread of security breaches and provides a secure platform for running diverse workloads.

5.7 Disaster Recovery and High Availability:

Virtualization improves the disaster recovery capabilities and high availability of cloud services. Virtual machines can be replicated and backed up more easily, allowing for faster recovery in the event of hardware failures or system disruptions. VM migration and failover capabilities enable seamless service continuity and minimize downtime.

5.8 Cost Efficiency:

Virtualization contributes to cost efficiency in cloud computing. Organizations can reduce hardware and energy costs by consolidating multiple VMs on a smaller number of physical servers. Virtualization also simplifies capacity planning and allows for efficient use of resources, resulting in cost savings and improved ROI.

CHAPTER – 6

Virtualization management and tools

Virtualization management tools are designed to facilitate the administration, monitoring, and control of virtualized environments. These tools provide a range of functionalities to streamline virtual infrastructure management, optimize resource utilization, ensure performance and availability, and simplify day-to-day administrative tasks. Here are some commonly used virtualizations management tools:



Figure 6.1 Tools

6.1 VMware vSphere:

VMware vSphere is a comprehensive virtualization platform that includes various management tools. It provides centralized management of virtual machines, storage, and networking resources. vSphere features functionalities like VM provisioning, migration, resource allocation, high availability, and fault tolerance. It also offers advanced capabilities such as distributed resource scheduling, virtual machine replication, and disaster recovery.

6.2 Microsoft System Centre Virtual Machine Manager (SCVMM):

SCVMM is a Microsoft Hyper-V virtualization platform management tool. It offers centralized management of virtual machines, virtual networking, and storage. SCVMM allows administrators to create, provision, and manage VMs, monitor performance, and automate resource allocation. It integrates with other Microsoft System Centre components to provide comprehensive management capabilities.

6.3 Citrix XenCenter:

XenCenter is a management console for Citrix XenServer, an open-source virtualization platform. It provides a user-friendly interface to manage virtual machines, storage, and networking resources. XenCenter allows administrators to create, clone, migrate, and monitor VMs, as well as manage resource pools and implement high-availability features.

6.4 Red Hat Virtualization Manager:

Red Hat Virtualization Manager is a management platform for Red Hat Virtualization, based on the open-source KVM hypervisor. It offers centralized administration of virtualized infrastructure, including VM provisioning, monitoring, and resource allocation. Red Hat Virtualization Manager supports features like live migration, high availability, and backup and recovery.

6.5 Proxmox Virtual Environment (Proxmox VE):

Proxmox VE is an open-source virtualization management platform that combines KVM and LXC containerization. It provides a web-based management interface for creating, managing, and monitoring virtual machines and containers. Proxmox VE offers features like live migration, high availability, backup, and clustering.

6.6 OpenStack:

OpenStack is an open-source cloud computing platform that includes virtualization management capabilities. It provides a framework for managing computing, storage, and networking resources in a cloud environment. OpenStack supports multiple hypervisors, including KVM, VMware, and Hyper-V, and offers a range of management tools and APIs for provisioning and monitoring virtual instances.

6.7 Docker and Kubernetes:

Docker and Kubernetes are popular containerization platforms that include management tools for deploying and managing containerized applications. While not strictly virtualization in the traditional sense, these platforms provide efficient resource utilization and isolation for running applications, along with robust management capabilities for container orchestration and scaling.

CHAPTER - 7

Use cases and success stories in virtualization

Virtualization has been widely adopted across various industries and has proven to be beneficial in numerous use cases. Here are some notable use cases and success stories that demonstrate the impact and advantages of virtualization:



Figure 7.1 Success

7.1 Server Virtualization for Data Centres:

One of the most common use cases for virtualization is server virtualization in data centers. By consolidating multiple physical servers into virtual machines running on a single physical host, organizations can optimize resource utilization, reduce hardware costs, and improve scalability and flexibility. VMware is a prominent example of a company that has successfully implemented server virtualization solutions.

7.2 Desktop Virtualization:

Virtualizing desktop environments allow users to access their desktops and applications from any device or location. This approach enhances mobility, security, and centralized management of desktop environments. Citrix XenDesktop and VMware Horizon have widely adopted desktop virtualization solutions, enabling organizations to efficiently deliver virtual desktops and applications to end users.

7.3 Application Virtualization:

Application virtualization enables applications to run in isolated environments, decoupling them from the underlying operating system and hardware. This approach simplifies application deployment, improves compatibility, and enhances security by isolating applications from each other. Microsoft App-V and VMware ThinApp are examples of successful application

virtualization solutions.

7.4 Network Function Virtualization (NFV):

NFV involves virtualizing network functions such as firewalls, routers, and load balancers, replacing dedicated hardware appliances with software-based virtual instances. NFV allows for greater flexibility, scalability, and cost savings in network infrastructure. AT&T is a notable example of a telecommunications company that has successfully implemented NFV to transform its network architecture.

7.5 Cloud Computing:

Virtualization is a fundamental technology underlying cloud computing. Cloud service providers leverage virtualization to create and manage virtualized resources across the country provisioned and scaled based on customer demand. Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform are leading cloud providers that heavily rely on virtualization to deliver scalable and cost-effective cloud services.

7.6 Disaster Recovery and Business Continuity:

Virtualization plays a crucial role in enabling efficient disaster recovery and business continuity solutions. By replicating virtual machines to a secondary site or cloud environment, organizations can quickly recover their critical systems and data in the event of a disaster. VMware Site Recovery Manager (SRM) and Zerto are widely used virtualization-based disaster recovery solutions.

7.7 Testing and Development Environments:

Virtualization provides an ideal platform for creating and managing testing and development environments. By utilizing virtual machines, developers and testers can rapidly provision and replicate different configurations, isolate test environments, and reduce resource requirements. This improves efficiency and enables faster software development cycles.

These are just a few examples of the many successful use cases and implementations of virtualization. The versatility and benefits of virtualization have made it a fundamental technology in modern IT infrastructure, driving innovation, efficiency, and cost savings across various industries.

CHAPTER - 8

Future Trends and Emerging Technologies

Virtualization continues to evolve and adapt to the changing landscape of technology. Here are some future trends and emerging technologies that are shaping the future of virtualization:

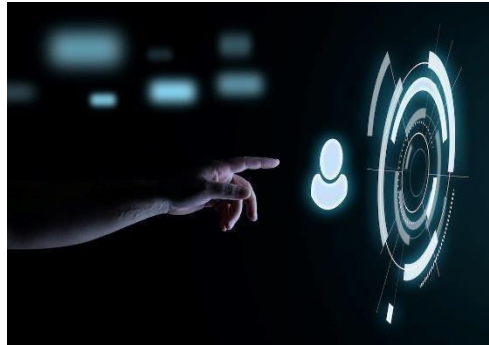


Figure 8.1 Future

8.1 Containerization:

Containerization technology, led by platforms like Docker and Kubernetes, is gaining popularity as an alternative to traditional virtualization. Containers offer lightweight, isolated environments for running applications, allowing for greater efficiency, faster startup times, and improved scalability. The integration of containerization with virtualization technologies is becoming more prevalent, enabling organizations to leverage the benefits of both approaches.

8.2 Edge Computing and Virtualization:

The rise of edge computing, where computing resources are located closer to the data source or end-user, presents new opportunities for virtualization. Virtualization at the edge allows for the deployment and management of virtualized environments in edge devices, enabling efficient resource utilization, data processing, and real-time decision-making. This trend is particularly relevant in IoT (Internet of Things) scenarios and applications that require low latency.

8.3 Serverless Computing:

Serverless computing, also known as Function as a Service (FaaS), abstracts away the need to manage infrastructure resources and focuses on running individual functions or tasks. Virtualization plays a crucial role in serverless platforms by dynamically provisioning and managing the underlying resources needed to execute functions. This technology allows organizations to focus on writing code without the need for infrastructure management.

8.4 AI and Machine Learning in Virtualization:

Artificial Intelligence (AI) and Machine Learning (ML) techniques are being leveraged to optimize virtualization and resource management. AI-powered algorithms can analyze workload patterns, resource usage, and performance data to dynamically allocate and optimize resources based on demand, leading to improved efficiency, cost savings, and better performance in virtualized environments.

8.5 Network Virtualization:

Network virtualization is gaining momentum as organizations seek to abstract network functions from physical hardware and create virtual networks that can be provisioned, configured, and managed programmatically. Software-defined networking (SDN) and network function virtualization (NFV) technologies are key enablers of network virtualization, providing flexibility, agility, and scalability in network infrastructure management.

8.6 Security and Virtualization:

As virtualized environments become more prevalent, ensuring security becomes crucial. Emerging technologies are focusing on enhancing the security of virtualized infrastructure, such as virtual machine isolation, encryption of VM data, and secure VM migration techniques. Innovations in virtualization security aim to protect against vulnerabilities, data breaches, and unauthorized access in virtualized environments.

8.7 Hybrid and Multi-Cloud Virtualization:

Organizations are increasingly adopting hybrid and multi-cloud strategies, combining on-premises infrastructure with public and private cloud resources. Virtualization plays a vital role in enabling workload portability, seamless migration, and resource management across heterogeneous cloud environments. Technologies like VMware Cloud Foundation and Azure Arc are emerging to provide unified management and virtualization capabilities across different cloud platfor

CHAPTER- 9

My Project

PROJECT: 01

My Project on virtualization is a Python program that makes an object drag and drop through hand gestures by handling the camera portion.



Figure 9.1 Drag and Drop

9.1 Program:

Python version: 3.10.11

Input:

```
import cv2

import mediapipe as mp
cap = cv2.VideoCapture(0)
mpHands = mp.solutions.hands
hands = mpHands.Hands()
mpDraw = mp.solutions.drawing_utils
red = (0, 0, 255)
green = (0, 255, 0)
centerPoints = [[120, 100], [300, 160], [80, 200]]
```

```

circleColors = [red, red, red]
inside = False
radius, color, thickness = 60, red, 20
while True:
    success, img = cap.read()
    img = cv2.flip(img, 1)
    imgRGB = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
    results = hands.process(imgRGB)
    multiLandMarks = results.multi_hand_landmarks
    if multiLandMarks:
        handPoints = []
        for handLms in multiLandMarks:
            mpDraw.draw_landmarks(img, handLms, mpHands.HAND_CONNECTIONS)
            for idx, lm in enumerate(handLms.landmark):
                # print(idx,lm)
                h, w, c = img.shape
                cx, cy = int(lm.x * w), int(lm.y * h)
                handPoints.append((cx, cy))
        for point in handPoints:
            cv2.circle(img, point, 10, (0, 0, 255), cv2.FILLED)
        firstFingerOpen = False
        secondFingerOpen = False
        if handPoints[8][1] < handPoints[6][1]:
            firstFingerOpen = True
        if handPoints[12][1] < handPoints[10][1]:
            secondFingerOpen = True
        for i in range(len(centerPoints)):
            d = int(radius ** 2 - ((centerPoints[i][0] - handPoints[8][0]) ** 2 + (centerPoints[i][1] -
handPoints[8][1]) ** 2))
            if d >= 0:
                inside = True

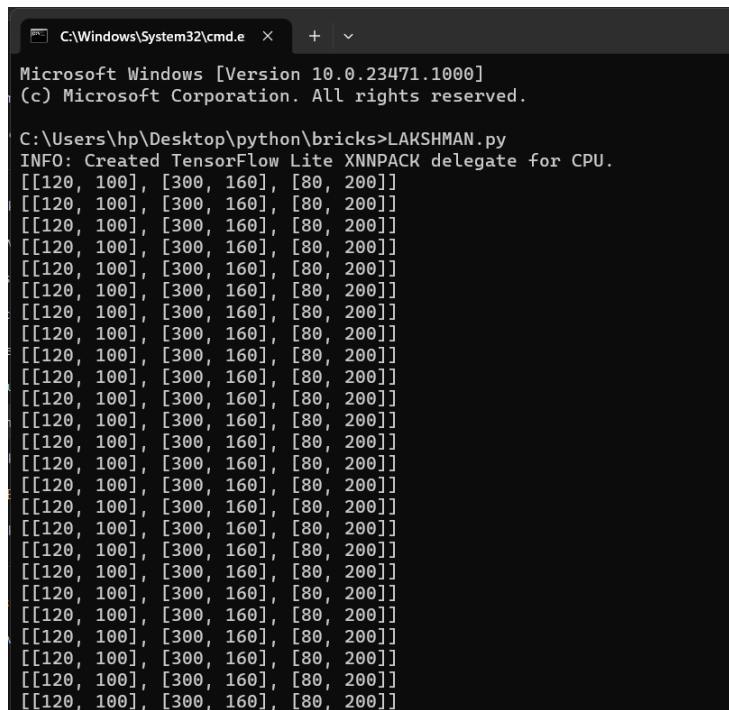
```

```

else:
    inside = False
if inside:
    if firstFingerOpen and secondFingerOpen:
        circleColors[i] = green
centerPoints[i] = handPoints[8]
brea
else:
    circleColors[i] = red
for point, color in zip(centerPoints, circleColors):
    cv2.circle(img, point, radius, color, thickness)
print(centerPoints)
cv2.imshow("Finger Counter", img)
cv2.waitKey(1)

```

Compiling the program:



```

C:\Windows\System32\cmd.e
Microsoft Windows [Version 10.0.23471.1000]
(c) Microsoft Corporation. All rights reserved.

C:\Users\hp\Desktop\python\bricks>LAKSHMAN.py
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
[[120, 100], [300, 160], [80, 200]]
[[120, 100], [300, 160], [80, 200]]
[[120, 100], [300, 160], [80, 200]]
[[120, 100], [300, 160], [80, 200]]
[[120, 100], [300, 160], [80, 200]]
[[120, 100], [300, 160], [80, 200]]
[[120, 100], [300, 160], [80, 200]]
[[120, 100], [300, 160], [80, 200]]
[[120, 100], [300, 160], [80, 200]]
[[120, 100], [300, 160], [80, 200]]
[[120, 100], [300, 160], [80, 200]]
[[120, 100], [300, 160], [80, 200]]
[[120, 100], [300, 160], [80, 200]]
[[120, 100], [300, 160], [80, 200]]
[[120, 100], [300, 160], [80, 200]]
[[120, 100], [300, 160], [80, 200]]
[[120, 100], [300, 160], [80, 200]]
[[120, 100], [300, 160], [80, 200]]
[[120, 100], [300, 160], [80, 200]]
[[120, 100], [300, 160], [80, 200]]

```

Figure 9.2 Compiling program

Output:

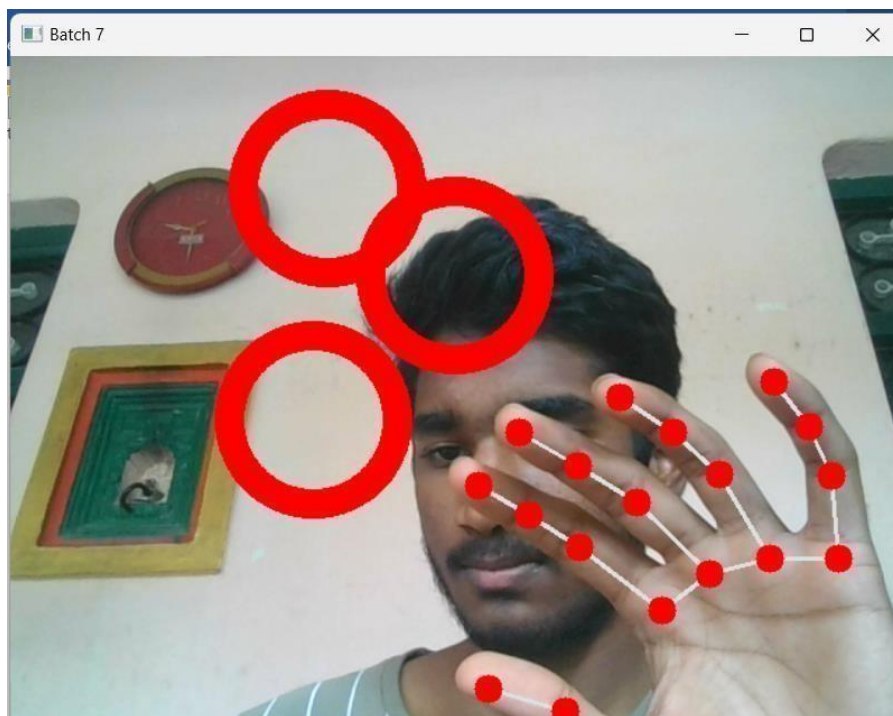


Figure 9.3 Output 1

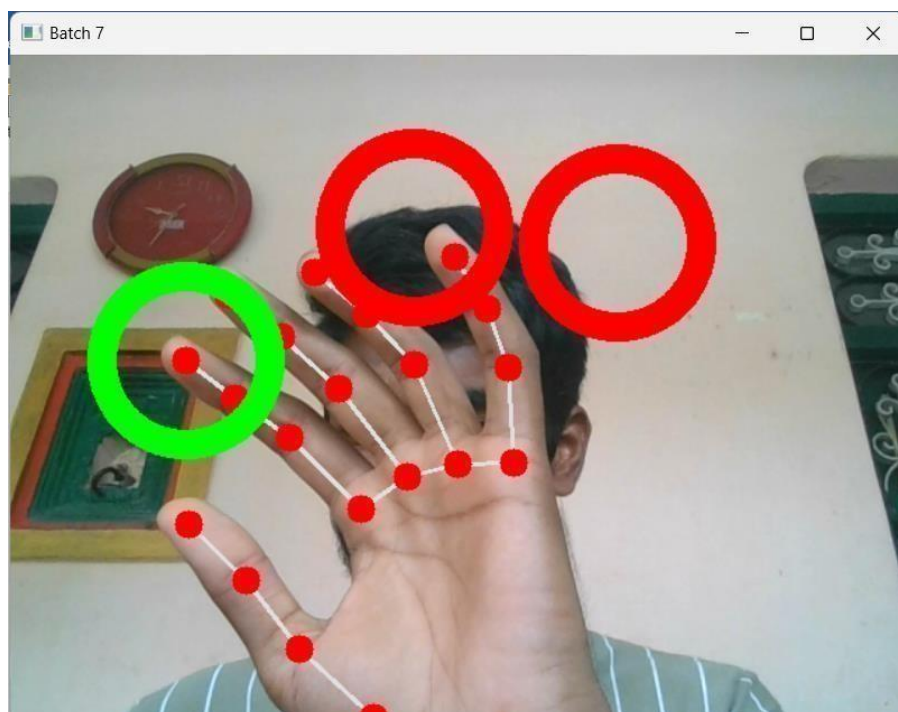


Figure 9.4 Output 2

9.2 About the program:

1. The program is a Python script that utilizes the OpenCV and Mediapipe libraries to track and visualize hand movements in real-time. It captures frames from a camera, detects hand landmarks, and provides interactive feedback based on the position and movements of the user's hand.
2. The script starts by initializing the video capture object to access the camera feed. It then sets up the hand-tracking module from the Mediapipe library, which allows it to detect and track hand landmarks in the captured frames.
3. Inside the main loop, the script processes each frame by detecting hand landmarks, drawing them on the frame, and analyzing finger movements. It determines if the user's fingers are open or closed and checks if they are interacting with virtual circles on the screen.
4. The script uses color-coded circles to represent the virtual interaction points. When the user's hand interacts with a circle by placing a fingertip inside it and having the right finger positions, the color of the circle changes to green, indicating successful interaction.
5. The output is displayed in a window named "Finger Counter" using OpenCV. The program continuously tracks the user's hand movements, providing visual feedback and demonstrating the capabilities of hand tracking and interaction in a virtual environment.
6. This program can serve as a starting point for various interactive applications, such as virtual reality experiences, gesture-based control systems, or interactive games, where hand movements play a crucial role.

9.3 Detailed Explanation of the Program:

1. **Importing Libraries:** The script starts by importing the necessary libraries, including cv2 for computer vision operations and mediapipe for hand tracking.
 2. **Setting up the Video Capture:** The script initializes the video capture object (cv2.VideoCapture) to access the default camera (index 0). This allows the script to capture frames from the camera for processing.
- Initializing Hand Tracking:** The script sets up the hand tracking module from the Mediapipe library. It creates an instance of the hands. Hands class, which provides functionality for detecting and tracking hand landmarks

3. **Defining Variables and Colors:** The script defines variables for center points, circle colors, and other parameters used for visualization, such as the radius and thickness of the circles. It also defines red and green colors.
4. **Main Loop:** The script enters a while loop, where it continuously captures frames from the video stream, processes them, and displays the output.
5. **Hand Landmark Detection:** Inside the loop, the script reads a frame from the video capture, flips it horizontally (`cv2.flip`), and converts it to the RGB color space (`cv2.cvtColor`). It then passes the RGB frame to the hand-tracking module for hand landmark detection (`hands process`).
6. **Processing Hand Landmarks:** If hand landmarks are detected in the frame (`multiLandMarks`), the script iterates over each hand landmark and extracts the coordinates (`cx`, `cy`). It draws the landmarks on the frame using `mpDraw.draw_landmarks`.
7. **Finger Detection:** The script determines the openness of the first and second fingers by comparing the y-coordinates of specific hand landmarks.
8. **Circle Interaction:** The script calculates the distance between the center points and the fingertip of the hand, and based on the distance, determines if the fingertip is inside a circle or not. If the fingertip is inside a circle and both the first and second fingers are open, the color of the circle changes to green, indicating interaction.
9. **Displaying the Output:** The script displays the annotated frame with circles and landmarks using `cv2.imshow`. It waits for a key press (`cv2.waitKey`) with a delay of 1 millisecond before moving to the next iteration of the loop.

The script continuously tracks the user's hand, detects finger movements, and changes the color of the circles based on interaction. It provides a visual representation of hand tracking and interaction with virtual circles on the screen.

9.4 How To Run the Program:

To run the above program, you need to follow these steps:

1. **Set up the Python environment:** Make sure you have Python installed on your system. You can download and install Python from the official Python website (<https://www.python.org>).
2. **Install the required libraries:** OpenCV and Mediapipe are required for this program. You can install them using pip, the package manager for Python. Open a command prompt or terminal and run the following commands:

```
pip install opencv-python
```

```
pip install mediapipe
```

3. Copy the code: Copy the entire code provided in your question and save it as a Python file (e.g., `finger_counter.py`).
4. Connect a webcam: Ensure that you have a webcam connected to your computer, as the program will capture frames from the webcam.
5. Run the program: Open a command prompt or terminal, navigate to the directory where you saved the Python file, and execute the following command:

```
python finger_counter.py
```

6. Observe the output: The program will open a window displaying the webcam feed. Place your hand in front of the webcam, and the program will track your finger positions and update the circle colors accordingly.
7. Terminate the program: To stop the program, simply close the window displaying the webcam feed or press Ctrl+C in the command prompt or terminal where the program is running.

Make sure you have the necessary dependencies installed, the webcam is properly connected, and the code is saved as a Python file. Following these steps should allow you to run the finger counter program successfully

PROJECT:02

Virtual steering refers to the simulation of steering a vehicle in a digital or virtual environment, typically using input devices and software to replicate the experience. It is commonly used in video games, virtual reality simulations, and driving training programs to provide users with an immersive and realistic driving experience.



Figure:9.5 Virtual Steering

9.5 Program:

Key input:

```
import ctypes
```

```
keys = {  
    "w":0x11,  
    "a":0x1E,  
    "s":0x1F,  
    "d":0x20,  
}
```

```
PUL = ctypes.POINTER(ctypes.c_ulong)
```

```
class KeyBdInput(ctypes.Structure):  
    _fields_ = [("wVk", ctypes.c_ushort),  
                ("wScan", ctypes.c_ushort),  
                ("dwFlags", ctypes.c_ulong),  
                ("time", ctypes.c_ulong),  
                ("dwExtraInfo", PUL)]
```

```
class HardwareInput(ctypes.Structure):  
    _fields_ = [("uMsg", ctypes.c_ulong), ("wParamL",  
                ctypes.c_short), ("wParamH",  
                ctypes.c_ushort)]
```

```
class MouseInput(ctypes.Structure):
```

```

_fields_ = [("dx", ctypes.c_long), ("dy",
    ctypes.c_long), ("mouseData",
    ctypes.c_ulong), ("dwFlags",
    ctypes.c_ulong),
    ("time", ctypes.c_ulong),
    ("dwExtraInfo", PUL)]

class Input_I(ctypes.Union):
    _fields_ = [("ki", KeyBdInput),
        ("mi", MouseInput),
        ("hi", HardwareInput)]

class Input(ctypes.Structure):
    _fields_ = [("type", ctypes.c_ulong),
        ("ii", Input_I)]

def press_key(key):
    extra = ctypes.c_ulong(0)
    ii_ = Input_I()
    ii_.ki = KeyBdInput( 0, keys[key], 0x0008, 0, ctypes.pointer(extra) )x =
    Input( ctypes.c_ulong(1), ii_ )
    ctypes.windll.user32.SendInput(1, ctypes.pointer(x), ctypes.sizeof(x))

def release_key(key):
    extra = ctypes.c_ulong(0)
    ii_ = Input_I()
    ii_.ki = KeyBdInput( 0, keys[key], 0x0008 | 0x0002, 0, ctypes.pointer(extra) )x =
    Input( ctypes.c_ulong(1), ii_ )
    ctypes.windll.user32.SendInput(1, ctypes.pointer(x), ctypes.sizeof(x))

```

Steering:

```

import math
import keyinput
import cv2
import mediapipe as mp
mp_drawing = mp.solutions.drawing_utils
mp_drawing_styles = mp.solutions.drawing_styles
mp_hands = mp.solutions.hands
font = cv2.FONT_HERSHEY_SIMPLEX
# 0 For webcam input:
cap = cv2.VideoCapture(0)

with mp_hands.Hands( model_complexity=0,
    min_detection_confidence=0.5,
    min_tracking_confidence=0.5) as hands:

```

```

while cap.isOpened():
    success, image = cap.read()
    if not success:
        print("Ignoring empty camera frame.")
        # If loading a video, use 'break' instead of 'continue'.
        continue

    # To improve performance, optionally mark the image as not writeable to
    image.flags.writeable = False
    image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
    results = hands.process(image)
    imageHeight, imageWidth, _ = image.shape

    # Draw the hand annotations on the image.
    image.flags.writeable = True
    image = cv2.cvtColor(image, cv2.COLOR_RGB2BGR)
    co=[]
    if results.multi_hand_landmarks:
        for hand_landmarks in results.multi_hand_landmarks:
            mp_drawing.draw_landmarks(
                image,
                hand_landmarks,
                mp_hands.HAND_CONNECTIONS,
                mp_drawing_styles.get_default_hand_landmarks_style(),
                mp_drawing_styles.get_default_hand_connections_style())
            for point in mp_hands.HandLandmark:
                if str(point) == "HandLandmark.WRIST":
                    normalizedLandmark = hand_landmarks.landmark[point]
                    pixelCoordinatesLandmark =
mp_drawing._normalized_to_pixel_coordinates(normalizedLandmark.x,
                                             normalizedLandmark.y,
                                             imageWidth, imageHeight)

                try:
                    co.append(list(pixelCoordinatesLandmark))
                except:
                    continue

    if len(co) == 2:
        xm, ym = (co[0][0] + co[1][0]) / 2, (co[0][1] + co[1][1]) / 2
        radius = 150
        try:
            m=(co[1][1]-co[0][1])/(co[1][0]-co[0][0])

```

```

except:
    continue
a = 1 + m ** 2
b = -2 * xm - 2 * co[0][0] * (m ** 2) + 2 * m * co[0][1] - 2 * m * ym
c = xm ** 2 + (m ** 2) * (co[0][0] ** 2) + co[0][1] ** 2 + ym ** 2 - 2 * co[0][1] * ym -
2 * co[0][1] * co[0][
    0] * m + 2 * m * ym * co[0][0] - 22500
xa = (-b + (b ** 2 - 4 * a * c) ** 0.5) / (2 * a)xb
    = (-b - (b ** 2 - 4 * a * c) ** 0.5) / (2 * a)
ya = m * (xa - co[0][0]) + co[0][1]
yb = m * (xb - co[0][0]) + co[0][1]if
m!=0:
    ap = 1 + ((-1/m) ** 2)
    bp = -2 * xm - 2 * xm * ((-1/m) ** 2) + 2 * (-1/m) * ym - 2 * (-1/m) * ym
    cp = xm ** 2 + ((-1/m) ** 2) * (xm ** 2) + ym ** 2 + ym ** 2 - 2 * ym * ym - 2 *ym *
xm * (-1/m) + 2 * (-1/m) * ym * xm - 22500
    try:
        xap = (-bp + (bp ** 2 - 4 * ap * cp) ** 0.5) / (2 * ap)xbp
        = (-bp - (bp ** 2 - 4 * ap * cp) ** 0.5) / (2 * ap) yap = (-
        1 / m) * (xap - xm) + ym
        ybp = (-1 / m) * (xbp - xm) + ym

except:
    continue

cv2.circle(img=image, center=(int(xm), int(ym)), radius=radius, color=(195, 255, 62),
thickness=15)

l = (int(math.sqrt((co[0][0] - co[1][0]) ** 2 * (co[0][1] - co[1][1]) ** 2)) - 150) // 2
cv2.line(image, (int(xa), int(ya)), (int(xb), int(yb)), (195, 255, 62), 20)
if co[0][0] > co[1][0] and co[0][1]>co[1][1] and co[0][1] - co[1][1] > 65:
    print("Turn left.")
    keyinput.release_key('s')
    keyinput.release_key('d')
    keyinput.press_key('a')
    cv2.putText(image, "Turn left", (50, 50), font, 0.8, (0, 255, 0), 2, cv2.LINE_AA)
    cv2.line(image, (int(xbp), int(ybp)), (int(xm), int(ym)), (195, 255, 62), 20)

elif co[1][0] > co[0][0] and co[1][1]> co[0][1] and co[1][1] - co[0][1] > 65:
    print("Turn left.")
    keyinput.release_key('s')
    keyinput.release_key('d')
    keyinput.press_key('a')

```

```
cv2.putText(image, "Turn left", (50, 50), font, 0.8, (0, 255, 0), 2, cv2.LINE_AA)
cv2.line(image, (int(xbp), int(ybp)), (int(xm), int(ym)), (195, 255, 62), 20)
```

```
elif co[0][0] > co[1][0] and co[1][1] > co[0][1] and co[1][1] - co[0][1] > 65:
```

```
    print("Turn right.")
    keyinput.release_key('s')
    keyinput.release_key('a')
    keyinput.press_key('d')
    cv2.putText(image, "Turn right", (50, 50), font, 0.8, (0, 255, 0), 2, cv2.LINE_AA)
    cv2.line(image, (int(xap), int(yap)), (int(xm), int(ym)), (195, 255, 62), 20)
```

```
elif co[1][0] > co[0][0] and co[0][1] > co[1][1] and co[0][1] - co[1][1] > 65:
```

```
    print("Turn right.")
    keyinput.release_key('s')
    keyinput.release_key('a')
    keyinput.press_key('d')
    cv2.putText(image, "Turn right", (50, 50), font, 0.8, (0, 255, 0), 2, cv2.LINE_AA)
    cv2.line(image, (int(xap), int(yap)), (int(xm), int(ym)), (195, 255, 62), 20)
```

```
else:
```

```
    print("keeping straight")
    keyinput.release_key('s')
    keyinput.release_key('a')
    keyinput.release_key('d')
    keyinput.press_key('w')
    cv2.putText(image, "keep straight", (50, 50), font, 0.8, (0, 255, 0), 2,
```

```
cv2.LINE_AA)
```

```
    if ybp > yap:
```

```
        cv2.line(image, (int(xbp), int(ybp)), (int(xm), int(ym)), (195, 255, 62), 20)
    else:
        cv2.line(image, (int(xap), int(yap)), (int(xm), int(ym)), (195, 255, 62), 20)
```

```
if len(co) == 1: print("keeping back")
```

```
    keyinput.release_key('a')
    keyinput.release_key('d')
    keyinput.release_key('w')
    keyinput.press_key('s')
    cv2.putText(image, "keeping back", (50, 50), font, 1.0, (0, 255, 0), 2, cv2.LINE_AA)
```

```
cv2.imshow('MediaPipe Hands', cv2.flip(image, 1))
```

```
# Flip the image horizontally for a selfie-view display.
```

```
cv2.waitKey(5) & 0xFF == ord('q'):
    break
cap.release()
```

9.6 Explanation of the above programs:

Certainly! The provided Python script is a program that utilizes the MediaPipe library to track hand landmarks and interpret hand movements for virtual steering control. Here's a detailed explanation of how the program works:

1. Importing Libraries:

- The program starts by importing necessary libraries such as ``math``, ``keyinput``, ``cv2`` (OpenCV), ``mediapipe``, and related modules for hand tracking.

2. Initializing Variables and Webcam Capture:

- Various variables, including font type and webcam capture, are initialized.
- The ``cap`` variable is set to capture video from the default webcam (index 0).

3. Hand Tracking Loop:

- The main logic of the program lies within this loop, which runs until the webcam capture is open.
- The loop performs the following steps for each frame captured from the webcam:

4. Reading and Processing the Webcam Frame:

- The program reads a frame from the webcam using ``cap.read()``.
- If no frame is captured, the loop continues to the next iteration.

5. Hand Landmark Detection:

- The captured frame is converted to RGB format and processed by the ``hands.process()`` method to detect hand landmarks.
- The ``results`` object contains the detected hand landmarks and related information.

6. Drawing Hand Landmarks:

- The program retrieves the image dimensions and converts the frame back to BGR format for drawing.
- It uses ``mp_drawing.draw_landmarks`` to draw the detected hand landmarks on the frame, connecting them with lines.
- The coordinates of the wrist landmarks are extracted and stored in the ``co`` list.

7. Calculating Steering:

- If two hand landmarks (wrist landmarks) are detected (``len(co) == 2``), the program calculates the steering action.
- It calculates the midpoint (``xm``, ``ym``) between the two wrist landmarks and defines a radius

for the steering wheel.

- The program calculates the slope (m) of the line connecting the wrist landmarks.
- Using the slope and midpoint, it calculates the intersection points (x_a , y_a) and (x_b , y_b) between the steering wheel and the line.

8. Determining Steering Action:

- Based on the relative positions and distances between the wrist landmarks and the calculated intersection points, the program determines the steering action:
 - If the left turn conditions are met, it simulates a left turn by pressing the 'a' key and displays "Turn left" on the frame.
 - If the right turn conditions are met, it simulates a right turn by pressing the 'd' key and displays "Turn right" on the frame.
 - If none of the above conditions are met, it simulates driving straight by pressing the 'w' key and displays "Keep straight" on the frame.

9. Simulating Moving Backwards:

- If only one hand landmark is detected ($\text{len(co)} == 1$), the program simulates moving backwards by pressing the 's' key and displays "Keeping back" on the frame.

10. Displaying the Output:

- The program displays the annotated frame with the detected hand landmarks and the corresponding steering action using `cv2.imshow()`.

11. Exiting the Program:

- The program waits for a key press, and if the 'q' key is pressed, it breaks out of the loop.
- Finally, the webcam capture is released, and the program terminates.

Output:

```
Microsoft Windows [Version 10.0.23475.1000]
(c) Microsoft Corporation. All rights reserved.

C:\Users\hp\Desktop\VirtualSteering-main>keyinput.py

C:\Users\hp\Desktop\VirtualSteering-main>steering.py
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
keeping back
keeping back
keeping back
keeping back
keeping back
keeping back
keeping back
keeping straight
keeping straight
keeping back
keeping back
keeping back
Turn right.
Turn right.
Turn right.
Turn right.
Turn right.
keeping straight
keeping straight
keeping straight
keeping straight
keeping straight
Turn left.
```

Figure 9.6 Compiling Program 2

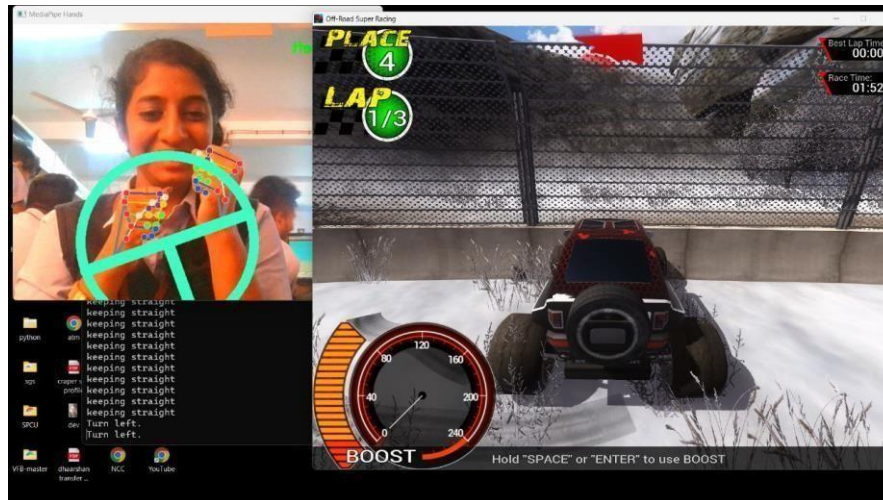


Figure: 9.7 Output 1

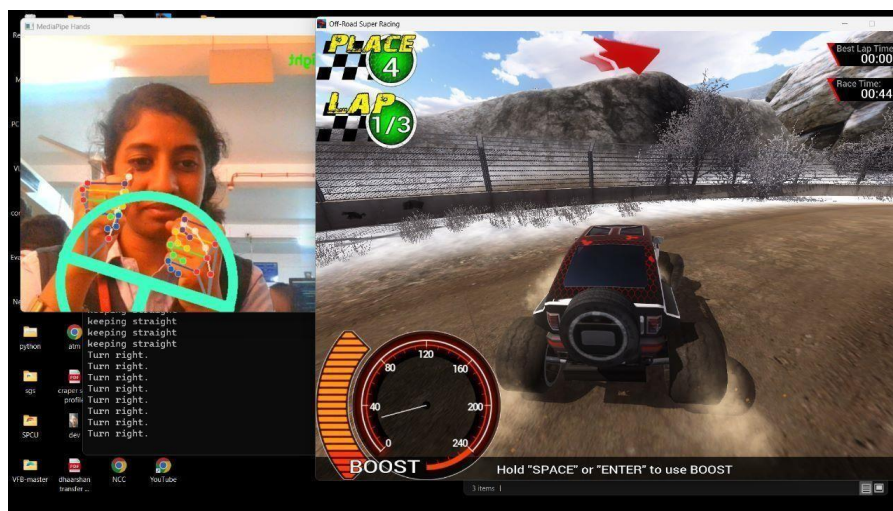


Figure:9.8 Output 2

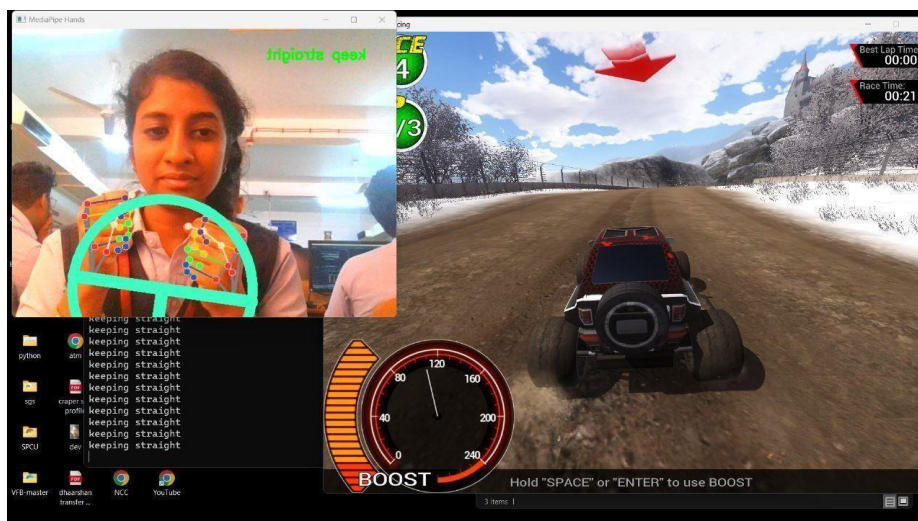


Figure:9.9 Output 3

10. CONCLUSION

Virtualization should never be seen as a simple solution to a specific problem; that is the main idea we have been trying to convey. It is a principle, a technology that is applicable in a very large range of different solutions. It is also a buzzword to get cash flowing nowadays, and is heralded by a lot of companies as "the next best thing in IT".

The purpose of this article is to make clear that we believe the current wave of virtualization will, in fact, make very big changes in data centers as we know them, but also that it is not some new "holy grail of IT" and has always been a part of it in some way. It's important not to get carried away by all the recent hype surrounding the subject, but to understand each technology's purpose and how to make use of them in the most efficient way possible.

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